

JSS 1-3

12-WEEK TEACHER LESSON PLAN

HYPOTHESIZE -> TEST -> QUANTIFY -> ANALYZE -> MANAGE

Structure	Weeks 1-10: one GrowMeal garden per week. Week 11: synthesis, remediation and portfolio. Week 12: celebration, demonstration and level completion.
Integrated resources	Vocabulary -> Flashcards -> Teacher Demonstration -> Practical Activity -> Journal -> Garden Mission -> Ingredient Lab/Farm-to-Food/Food Smart -> Assessment -> Passport.
Teacher preparation	Inspect the garden/site, choose exact cards/activities, prepare materials and DIY substitutes, review safety, assign learner groups and pre-check instruments.
Evidence rule	Every week should leave visible evidence: oral/drawn evidence at Nursery; records and measurements in Primary; increasingly traceable data, analysis and decisions at JSS/SSS.
Safety	School safeguarding, food-safety and tool-use rules take priority. No tasting is required. Teachers control knives, heat, chemicals, powered equipment and other higher-risk tools.

WEEK 01 - Production Garden

Learning objective	Conduct a structured investigation in the Production Garden, quantify a response and convert evidence into a management recommendation.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Flashcards for Production Garden; TD-301 to TD-310; PA-301 to PA-310; GM-301 to GM-310; AQ-301 to AQ-310.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-301–IL-310, FF-301–FF-310, or FSC-301–FSC-310 activity where it fits the week's garden evidence.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Use observation plus AQ-301–AQ-310; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 02 - Raised Systems Garden

Learning objective	Conduct a structured investigation in the Raised Systems Garden, quantify a response and convert evidence into a management recommendation.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Flashcards for Raised Systems Garden; TD-311 to TD-320; PA-311 to PA-320; GM-311 to GM-320; AQ-311 to AQ-320.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-311–IL-320, FF-311–FF-320, or FSC-311–FSC-320 activity where it fits the week's garden evidence.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Use observation plus AQ-311–AQ-320; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 03 - Grid Experiment Garden

Learning objective	Conduct a structured investigation in the Grid Experiment Garden, quantify a response and convert evidence into a management recommendation.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Flashcards for Grid Experiment Garden; TD-321 to TD-330; PA-321 to PA-330; GM-321 to GM-330; AQ-321 to AQ-330.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-321–IL-330, FF-321–FF-330, or FSC-321–FSC-330 activity where it fits the week's garden evidence.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Use observation plus AQ-321–AQ-330; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 04 - Crop Rotation Garden

Learning objective	Conduct a structured investigation in the Crop Rotation Garden, quantify a response and convert evidence into a management recommendation.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Flashcards for Crop Rotation Garden; TD-331 to TD-340; PA-331 to PA-340; GM-331 to GM-340; AQ-331 to AQ-340.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-331–IL-340, FF-331–FF-340, or FSC-331–FSC-340 activity where it fits the week's garden evidence.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Use observation plus AQ-331–AQ-340; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 05 - Intercropping Garden

Learning objective	Conduct a structured investigation in the Intercropping Garden, quantify a response and convert evidence into a management recommendation.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Flashcards for Intercropping Garden; TD-341 to TD-350; PA-341 to PA-350; GM-341 to GM-350; AQ-341 to AQ-350.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-341–IL-350, FF-341–FF-350, or FSC-341–FSC-350 activity where it fits the week's garden evidence.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Use observation plus AQ-341–AQ-350; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 06 - Water-Smart Garden

Learning objective	Conduct a structured investigation in the Water-Smart Garden, quantify a response and convert evidence into a management recommendation.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Flashcards for Water-Smart Garden; TD-351 to TD-360; PA-351 to PA-360; GM-351 to GM-360; AQ-351 to AQ-360.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-351–IL-360, FF-351–FF-360, or FSC-351–FSC-360 activity where it fits the week's garden evidence.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Use observation plus AQ-351–AQ-360; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 07 - Compost & Soil Garden

Learning objective	Conduct a structured investigation in the Compost & Soil Garden, quantify a response and convert evidence into a management recommendation.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Flashcards for Compost & Soil Garden; TD-361 to TD-370; PA-361 to PA-370; GM-361 to GM-370; AQ-361 to AQ-370.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-361–IL-370, FF-361–FF-370, or FSC-361–FSC-370 activity where it fits the week's garden evidence.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Use observation plus AQ-361–AQ-370; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 08 - Soilless Learning Garden

Learning objective	Conduct a structured investigation in the Soilless Learning Garden, quantify a response and convert evidence into a management recommendation.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Flashcards for Soilless Learning Garden; TD-371 to TD-380; PA-371 to PA-380; GM-371 to GM-380; AQ-371 to AQ-380.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-371–IL-380, FF-371–FF-380, or FSC-371–FSC-380 activity where it fits the week's garden evidence.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Use observation plus AQ-371–AQ-380; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 09 - Sensor-Ready Garden

Learning objective	Conduct a structured investigation in the Sensor-Ready Garden, quantify a response and convert evidence into a management recommendation.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Flashcards for Sensor-Ready Garden; TD-381 to TD-390; PA-381 to PA-390; GM-381 to GM-390; AQ-381 to AQ-390.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-381–IL-390, FF-381–FF-390, or FSC-381–FSC-390 activity where it fits the week's garden evidence.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Use observation plus AQ-381–AQ-390; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 10 - Applied Systems Garden

Learning objective	Conduct a structured investigation in the Applied Systems Garden, quantify a response and convert evidence into a management recommendation.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Flashcards for Applied Systems Garden; TD-391 to TD-400; PA-391 to PA-400; GM-391 to GM-400; AQ-391 to AQ-400.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Select one IL-391–IL-400, FF-391–FF-400, or FSC-391–FSC-400 activity where it fits the week's garden evidence.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Use observation plus AQ-391–AQ-400; record practical, data/investigation and mission problem-solving evidence in the Food Smart Passport.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 11 - Synthesis & Portfolio

Learning objective	Synthesize evidence from the ten gardens, revisit weak competencies, complete unfinished missions, and prepare a portfolio of the learner's strongest evidence.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Student Journal; Food Smart Passport; selected Practical Cards, Garden Missions, Ingredient Lab, Farm-to-Food and Food Smart Challenge evidence from Weeks 1-10.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Learners choose evidence that shows growth in knowledge, practical competence, investigation/data skills and mission-based problem solving.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Teacher moderation: review journal, passport, assessment results and practical evidence; reteach or reassess competencies not yet secure.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?

WEEK 12 - GrowMeal Celebration & Demonstration

Learning objective	Celebrate learning through a safe GrowMeal showcase in which learners communicate what they grew, observed, measured, investigated and solved.
Vocabulary	12-18 scientific and systems terms; students must use terminology precisely in method, results and recommendation.
Resources	Food Smart Passport; learner portfolio; garden displays; selected charts/models; certificates or school-approved recognition materials.
Materials / DIY	Paper data sheets, calibrated marked containers, locally fabricated sampling frames/stakes and reference measurements.
Suggested lesson flow	Question/hypothesis (10 min) -> protocol/QC (15 min) -> replicated data collection (35 min) -> analysis (20 min) -> management decision (10 min).
Journal / evidence	Learners record the age-appropriate observation, drawing, measurement, data table, graph, analysis, reflection or design evidence generated this week.
Integrated extension	Garden walk, learner demonstrations, evidence exhibition, Farm-to-Food/Ingredient Lab displays and age-appropriate challenge presentations.
Questions to ask	What is the hypothesis? What is the response variable? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the decision?
Assessment	Final teacher verification, learner reflection, level-completion record and next-level learning goals.
AgriShine task	End with one real garden-care, improvement, communication or problem-solving action that leaves the learning environment better documented or better managed.

Teacher weekly review: What was achieved? Who needs support? What materials/garden maintenance are needed before the next lesson?